KAIST

2026 Quantum Field Theory

Homework 1

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Exercise 3.1 (not a homework)

The Hamiltonian, equation of motion, and the canonical commutation relations are given by

$$H = \frac{1}{2}\int d^3\mathbf{x}\ \left[\pi^2 + (\nabla\phi)^2 + m^2\phi^2\right], \tag{1}$$

$$\dot{\phi} = i[H,\phi], \qquad \dot{\pi} = i[H,\pi], \tag{2}$$

$$[\phi(\mathbf{x}),\pi(\mathbf{y})] = i\delta^3(\mathbf{x}-\mathbf{y}), \qquad [\phi(\mathbf{x}),\phi(\mathbf{y})] = [\pi(\mathbf{x}),\pi(\mathbf{y})] = 0. \tag{3}$$

Commutator of ϕ and π with each term of the Hamiltonian is calculated as follows:

$$\begin{aligned}
[\pi^2(\mathbf{y}),\phi(\mathbf{x})] &= \pi(\mathbf{y})[\pi(\mathbf{y}),\phi(\mathbf{x})] + [\pi(\mathbf{y}),\phi(\mathbf{x})]\pi(\mathbf{y}) \\
&= -2i\delta^{(3)}(\mathbf{y}-\mathbf{x})\pi(\mathbf{y}), &(4)\\
[(\nabla\phi)^2(\mathbf{y}),\phi(\mathbf{x})] &= (\nabla\phi)(\mathbf{y})\cdot[(\nabla\phi)(\mathbf{y}),\phi(\mathbf{x})] + [(\nabla\phi)(\mathbf{y}),\phi(\mathbf{x})]\cdot(\nabla\phi)(\mathbf{y}) \\
&= (\nabla\phi)(\mathbf{y})\cdot\nabla_{\mathbf{y}}[\phi(\mathbf{y}),\phi(\mathbf{x})] + \nabla_{\mathbf{y}}[\phi(\mathbf{y}),\phi(\mathbf{x})]\cdot(\nabla\phi)(\mathbf{y}) \\
&= 0, &(5)\\
[\phi^2(\mathbf{y}),\phi(\mathbf{x})] &= \phi(\mathbf{y})[\phi(\mathbf{y}),\phi(\mathbf{x})] + [\phi(\mathbf{y}),\phi(\mathbf{x})]\phi(\mathbf{y}) = 0, &(6)\\
[\pi^2(\mathbf{y}),\pi(\mathbf{x})] &= \pi(\mathbf{y})[\pi(\mathbf{y}),\pi(\mathbf{x})] + [\pi(\mathbf{y}),\pi(\mathbf{x})]\pi(\mathbf{y}) = 0, &(7)\\
[(\nabla\phi)^2(\mathbf{y}),\pi(\mathbf{x})] &= (\nabla\phi)(\mathbf{y})\cdot[(\nabla\phi)(\mathbf{y}),\pi(\mathbf{x})] + [(\nabla\phi)(\mathbf{y}),\pi(\mathbf{x})]\cdot(\nabla\phi)(\mathbf{y}) \\
&= (\nabla\phi)(\mathbf{y})\cdot\nabla_{\mathbf{y}}[\phi(\mathbf{y}),\pi(\mathbf{x})] + \nabla_{\mathbf{y}}[\phi(\mathbf{y}),\pi(\mathbf{x})]\cdot(\nabla\phi)(\mathbf{y}) \\
&= i(\nabla\phi)(\mathbf{y})\cdot\nabla_{\mathbf{y}}\delta^{(3)}(\mathbf{y}-\mathbf{x}) + i\nabla_{\mathbf{y}}\delta^{(3)}(\mathbf{y}-\mathbf{x})\cdot(\nabla\phi)(\mathbf{y}) \\
&= 2i(\nabla\phi)(\mathbf{y})\cdot\nabla_{\mathbf{y}}\delta^{(3)}(\mathbf{y}-\mathbf{x}), &(8)\\
[\phi^2(\mathbf{y}),\pi(\mathbf{x})] &= \phi(\mathbf{y})[\phi(\mathbf{y}),\pi(\mathbf{x})] + [\phi(\mathbf{y}),\pi(\mathbf{x})]\phi(\mathbf{y}) = 2i\delta^{(3)}(\mathbf{y}-\mathbf{x})\phi(\mathbf{y}). &(9)
\end{aligned}$$

Hence, the time derivatives are

$$\begin{aligned}
\dot{\phi}(\mathbf{x}) &= i[H,\phi] \\
&= \frac{i}{2}\int d^3\mathbf{y}\ \left[-2i\pi(\mathbf{y})\delta^{(3)}(\mathbf{y}-\mathbf{x})\right] \\
&= \pi(\mathbf{x}), &(10)\\
\dot{\pi}(\mathbf{x}) &= i[H,\pi] \\
&= \frac{i}{2}\int d^3\mathbf{y}\ \left[2i(\nabla\phi)(\mathbf{y})\cdot\nabla_{\mathbf{y}}\delta^{(3)}(\mathbf{y}-\mathbf{x}) + 2im^2\phi(\mathbf{y})\delta^{(3)}(\mathbf{y}-\mathbf{x})\right] \\
&= (\nabla^2\phi)(\mathbf{x}) - m^2\phi(\mathbf{x}) &(11)
\end{aligned}$$

By combining them,

$$\ddot{\phi} = \nabla^2\phi - m^2\phi \iff (\partial^2 - m^2)\phi = 0 \tag{12}$$

Exercise 3.2 (not a homework)

The mode expansion of free scalar field ϕ is given by

$$\phi(x) = \int d\Pi_p \left(a_{\mathbf{p}} e^{ip\cdot x} + a_{\mathbf{p}}^\dagger e^{-ip\cdot x} \right), \tag{13}$$

Where $d\Pi_p = \frac{d^3\mathbf{p}}{(2\pi)^{3/2}\sqrt{2E_{\mathbf{p}}}}$

Using mode expansion, the commutator of $\phi(x)$ and $\phi(y)$ is calculated as follows:

$$\begin{aligned}
[\phi(x), \phi(y)] &= \int d\Pi_p d\Pi_q \left([a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] e^{ip\cdot x} e^{-iq\cdot y} + [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}] e^{-ip\cdot x} e^{iq\cdot y} \right) \\
&= \int d\Pi_p d\Pi_q \left[\delta^{(3)}(\mathbf{p}-\mathbf{q}) e^{ip\cdot x} e^{-iq\cdot y} - \delta^{(3)}(\mathbf{p}-\mathbf{q}) e^{-ip\cdot x} e^{iq\cdot y} \right] \\
&= \int \frac{d^3\mathbf{p}}{2E_{\mathbf{p}}(2\pi)^3} \left[e^{ip\cdot(x-y)} - e^{-ip\cdot(x-y)} \right]
\end{aligned} \tag{14}$$

Where $[a_{\mathbf{p}}, a_{\mathbf{q}}] = 0$, $[a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger] = 0$ and $[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = \delta^{(3)}(\mathbf{p}-\mathbf{q})$ are used.

If $x-y$ is spacelike (i.e. $(x-y)^2 > 0$), we can choose a reference frame where $\tilde{x}^0 = \tilde{y}^0$. Moreover, $p\cdot(x-y)$ and the measure $d^3\mathbf{p}/(2E_{\mathbf{p}}) = d^4p\,\delta(p^2+m^2)\Theta(p^0)$ remain invariant after the transformation. Hence, we can write

$$\begin{aligned}
[\phi(x), \phi(y)] &= \int \frac{d^3\mathbf{p}}{2E_{\mathbf{p}}(2\pi)^3} \left[e^{ip\cdot(x-y)} - e^{-ip\cdot(x-y)} \right] \\
&= \int \frac{d^3\tilde{\mathbf{p}}}{2E_{\tilde{\mathbf{p}}}(2\pi)^3} \left[e^{i\tilde{\mathbf{p}}\cdot(\tilde{\mathbf{x}}-\tilde{\mathbf{y}})} - e^{-i\tilde{\mathbf{p}}\cdot(\tilde{\mathbf{x}}-\tilde{\mathbf{y}})} \right]
\end{aligned} \tag{15}$$

By filping the sign of $\tilde{\mathbf{p}}$ in the second term, we get

$$\begin{aligned}
[\phi(x), \phi(y)] &= \int \frac{d^3\tilde{\mathbf{p}}}{2E_{\tilde{\mathbf{p}}}(2\pi)^3} e^{i\tilde{\mathbf{p}}\cdot(\tilde{\mathbf{x}}-\tilde{\mathbf{y}})} - \int \frac{d^3(-\tilde{\mathbf{p}})}{2E_{-\tilde{\mathbf{p}}}(2\pi)^3} e^{i(-\tilde{\mathbf{p}})\cdot(\tilde{\mathbf{x}}-\tilde{\mathbf{y}})} \\
&= 0
\end{aligned} \tag{16}$$

This is consistent with the causality of the theory.

$$[\phi(x), \phi(y)] = 0 \quad \text{for } (x-y)^2 > 0 \tag{17}$$

Problem 3.1

The mode expansion of free scalar field ϕ is given by

$$\phi(\mathbf{x},0) = \int d\Pi_p \left(a_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} + a_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} \right), \tag{18}$$

The vacuum expectation is

$$\langle 0|\phi(\mathbf{x},0)|0\rangle = \int d\Pi_p \langle 0|a_{\mathbf{p}}|0\rangle e^{i\mathbf{p}\cdot\mathbf{x}} + \langle 0|a_{\mathbf{p}}^\dagger|0\rangle e^{-i\mathbf{p}\cdot\mathbf{x}} = 0 \tag{19}$$

because $a_{\mathbf{p}}|0\rangle = 0$ and $\langle 0|a_{\mathbf{p}}^\dagger = 0$.

Therefore, the ground state expectation of $A(a)$ is

$$\langle 0|A(a)|0\rangle = \frac{1}{(a\sqrt{\pi})^3} \int d^3\mathbf{x}\, \langle 0|\phi(\mathbf{x},0)|0\rangle e^{-|\mathbf{x}|^2/a^2} = 0 \tag{20}$$

For the expectation of $\phi(\mathbf{x},0)\phi(\mathbf{y},0)$, the only term that contributes is $a_{\mathbf{p}} a_{\mathbf{q}}^\dagger$. Hence,

$$\begin{aligned}
\langle 0|\phi(\mathbf{x},0)\phi(\mathbf{y},0)|0\rangle &= \int d\Pi_p d\Pi_q \langle 0|a_{\mathbf{p}} a_{\mathbf{q}}^\dagger|0\rangle e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{q}\cdot\mathbf{y}} \\
&= \int d\Pi_p d\Pi_q \langle 0|a_{\mathbf{q}}^\dagger a_{\mathbf{p}} + [a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger]|0\rangle e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{q}\cdot\mathbf{y}} \\
&= \int d\Pi_p d\Pi_q \delta^{(3)}(\mathbf{p}-\mathbf{q}) e^{i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{q}\cdot\mathbf{y}} \\
&= \int d\Pi_p \frac{1}{(2\pi)^{3/2}\sqrt{2E_{\mathbf{p}}}} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \\
&= \int \frac{d^3\mathbf{p}}{2(2\pi)^3 E_{\mathbf{p}}} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}
\end{aligned} \tag{21}$$

The expectation of $A^2(a)$ is

$$\begin{aligned}
\langle 0|A^2(a)|0\rangle &= \frac{1}{(a\sqrt{\pi})^6} \int d^3\mathbf{x} d^3\mathbf{y}\, \langle 0|\phi(\mathbf{x},0)\phi(\mathbf{y},0)|0\rangle e^{-|\mathbf{x}|^2/a^2} e^{-|\mathbf{y}|^2/a^2} \\
&= \frac{1}{2(a\sqrt{\pi})^6(2\pi)^3} \int d^3\mathbf{x} d^3\mathbf{y} d^3\mathbf{p} \frac{1}{\sqrt{m^2+|\mathbf{p}|^2}} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} e^{-|\mathbf{x}|^2/a^2} e^{-|\mathbf{y}|^2/a^2} \\
&= \frac{1}{2(a\sqrt{\pi})^6(2\pi)^3} \int d^3\mathbf{p}\, \frac{e^{-a^2|\mathbf{p}|^2/2}}{\sqrt{m^2+|\mathbf{p}|^2}} \int d^3\mathbf{x}\, e^{-|\mathbf{x}-ia^2\mathbf{p}/2|^2/a^2} \int d^3\mathbf{y}\, e^{-|\mathbf{y}+ia^2\mathbf{p}/2|^2/a^2} \\
&= \frac{1}{2(2\pi)^3} \int d^3\mathbf{p}\, \frac{e^{-a^2|\mathbf{p}|^2/2}}{\sqrt{m^2+|\mathbf{p}|^2}} \\
&= \frac{1}{4\pi^2} \int_0^\infty dp\, \frac{p^2 e^{-a^2p^2/2}}{\sqrt{m^2+p^2}}
\end{aligned} \tag{22}$$

To get an expression with modified Bessel function, substitute $p = m\sinh t$ and $dp = m\cosh t dt$. Then,

$$\begin{aligned}
\int dp\, \frac{p^2 e^{-a^2p^2/2}}{\sqrt{m^2+p^2}} &= m^2 \int_0^\infty dt\, \sinh^2 t e^{-a^2m^2\sinh^2 t/2} \\
&= \frac{m^2 e^{\frac{a^2m^2}{4}}}{4} \int_0^\infty du\, (\cosh u - 1) e^{-(a^2m^2/4)\cosh u}
\end{aligned} \tag{23}$$

Where $u = 2t$. Then, this is the integral representation of the modified Bessel function of the second kind, $K_\nu(x) = \int_0^\infty d\tau\, e^{-x\cosh\tau}\cosh\nu\tau$. Hence, The expectation of $A^2(a)$ is

$$\langle 0|A^2(a)|0\rangle = \frac{m^2 e^{\frac{a^2m^2}{4}}}{16\pi^2}\left[K_1\left(\frac{a^2m^2}{4}\right) - K_0\left(\frac{a^2m^2}{4}\right)\right] \tag{24}$$

Since $\langle 0|A(a)|0\rangle = 0$, it is also the variance of $A(a)$.

$$\mathrm{Var}\, A(a) = \frac{m^2 e^{\frac{a^2m^2}{4}}}{16\pi^2}\left[K_1\left(\frac{a^2m^2}{4}\right) - K_0\left(\frac{a^2m^2}{4}\right)\right] \tag{25}$$

For $ma \ll 1$, $K_1(z) \simeq 1/z$ is dominant and $K_0(z) \simeq -\ln(z/2) - \gamma$ is subleading. since $e^z \simeq 1$,

$$\mathrm{Var}\, A(a) \simeq \frac{m^2}{16\pi^2}\frac{4}{a^2m^2} \simeq \frac{1}{4\pi^2a^2} \tag{26}$$

Hence, $\alpha = 1/4\pi^2$ and $\beta = -2$.

for $ma \gg 1$, $K_\nu(z) \simeq \sqrt{\pi/(2z)}\left(1 + (4\nu^2 - 1)/(8z)\right)$ and $K_1(z) - K_0(z) \simeq \sqrt{\pi/(2z)}e^{-z}/(2z)$.

Hence,

$$\mathrm{Var}\, A(a) \simeq \frac{1}{2(2\pi)^{3/2}ma^3} \tag{27}$$

Hence, $\alpha = 1/(2(2\pi)^{3/2}m)$ and $\beta = -3$.

For the short distance, the variance diverges as $1/a^2$. This means $\phi(x)$ can have arbitrarily large fluctuations at short distance. This can be a UV divergence in the theory.

For the long distance, it is modes times fluctuation per modes.

Problem 3.2

(a)

To have Lorentz invariance, all of the indices must be contracted via the metric tensor. Upto two derivatives, the candidates are:

$$\partial_\mu\phi^\dagger\partial^\mu\phi, \quad \partial_\mu\phi\partial^\mu\phi, \quad \partial_\mu\phi^\dagger\partial^\mu\phi^\dagger, \quad \phi^\dagger\phi, \quad \phi^2, \phi^{\dagger 2} \tag{28}$$

By the reality of the action and global $U(1)$ symmetry, the only term that can be included in the Lagrangian is $\partial_\mu\phi^\dagger\partial^\mu\phi$ and $\phi^\dagger\phi$.

to make the kinetic energy term to be positive, the Lagrangian of the simplest natural complex scalar field theory upto a multiplicative constant is

$$\mathcal{L} = -\partial_\mu\phi^\dagger\partial^\mu\phi - m^2\phi^\dagger\phi \tag{29}$$

(b)

The canonical momenta conjugate to ϕ and $\phi^\dagger$ are

$$\pi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi}^\dagger, \qquad \pi^\dagger = \frac{\partial\mathcal{L}}{\partial\dot{\phi}^\dagger} = \dot{\phi} \tag{30}$$

And, the equal-time canonical commutation relations are

$$[\phi(\mathbf{x},t),\pi(\mathbf{y},t)] = i\delta^{(3)}(\mathbf{x}-\mathbf{y}), \qquad [\phi^\dagger(\mathbf{x},t),\pi^\dagger(\mathbf{y},t)] = i\delta^{(3)}(\mathbf{x}-\mathbf{y}) \tag{31}$$

$$[\phi(\mathbf{x},t),\phi^\dagger(\mathbf{y},t)] = 0, \qquad [\pi(\mathbf{x},t),\pi^\dagger(\mathbf{y},t)] = 0 \tag{32}$$

$$[\phi(\mathbf{x},t),\pi^\dagger(\mathbf{y},t)] = 0, \qquad [\phi^\dagger(\mathbf{x},t),\pi(\mathbf{y},t)] = 0 \tag{33}$$

The Hamiltonian is

$$\begin{aligned} H &= \int d^3\mathbf{x}\,\left(\pi\dot{\phi} + \pi^\dagger\dot{\phi}^\dagger - \mathcal{L}\right) \\ &= \int d^3\mathbf{x}\,\left(\pi^\dagger\pi + \nabla\phi^\dagger\cdot\nabla\phi + m^2\phi^\dagger\phi\right) \end{aligned} \tag{34}$$

And the commutators of ϕ and π with the Hamiltonian are

$$\begin{aligned} [H,\phi(\mathbf{x})] &= \int d^3\mathbf{y}\,\left([\pi^\dagger\pi(\mathbf{y}),\phi(\mathbf{x})] + [\nabla\phi^\dagger\cdot\nabla\phi(\mathbf{y}),\phi(\mathbf{x})] + [m^2\phi^\dagger\phi(\mathbf{y}),\phi(\mathbf{x})]\right) \\ &= \int d^3\mathbf{y}\,\pi^\dagger(\mathbf{y})[\pi(\mathbf{y}),\phi(\mathbf{x})] \\ &= \int d^3\mathbf{y}\,[-i\delta^{(3)}(\mathbf{x}-\mathbf{y})]\pi^\dagger(\mathbf{y}) \qquad (35) \\ &= -i\pi^\dagger(\mathbf{x}), \qquad (36) \\ [H,\pi(\mathbf{x})] &= \int d^3\mathbf{y}\,\left([\pi^\dagger\pi(\mathbf{y}),\pi(\mathbf{x})] + [\nabla\phi^\dagger\cdot\nabla\phi(\mathbf{y}),\pi(\mathbf{x})] + [m^2\phi^\dagger\phi(\mathbf{y}),\pi(\mathbf{x})]\right) \\ &= \int d^3\mathbf{y}\,\nabla\phi^\dagger(\mathbf{y})\cdot\nabla_{\mathbf{y}}[\phi(\mathbf{y}),\pi(\mathbf{x})] + m^2\phi^\dagger(\mathbf{y})[\phi(\mathbf{y}),\pi(\mathbf{x})] \qquad (37) \\ &= -i\nabla^2\phi^\dagger(\mathbf{x}) + im^2\phi^\dagger(\mathbf{x}) \qquad (38) \end{aligned}$$

This gives the time derivatives of ϕ, $\phi^\dagger$, π, and $\pi^\dagger$:

$$\dot{\phi} = i[H, \phi] = \pi^\dagger, \qquad \dot{\phi}^\dagger = i[H, \phi^\dagger] = \pi, \tag{39}$$

$$\dot{\pi} = i[H, \pi] = \nabla^2\phi^\dagger - m^2\phi^\dagger, \qquad \dot{\pi}^\dagger = i[H, \pi^\dagger] = \nabla^2\phi - m^2\phi \tag{40}$$

By combining them, the equations of motion are

$$(\partial^2 - m^2)\phi = 0, \qquad (\partial^2 - m^2)\phi^\dagger = 0 \tag{41}$$

To obtain the mode expansion, fourier transform the fields:

$$\phi(x) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot\mathbf{x}} \phi_{\mathbf{p}}(t) \tag{42}$$

And the equation of motion gives:

$$\ddot{\phi}_{\mathbf{p}}(t) + E_{\mathbf{p}}^2 \phi_{\mathbf{p}}(t) = 0 \tag{43}$$

Where $E_p = \sqrt{|\mathbf{p}|^2 + m^2}$

Hence, the momentum mode is sum of positive and negative frequency modes:

$$\phi_{\mathbf{p}}(t) = \frac{(2\pi)^{3/2}}{\sqrt{2E_{\mathbf{p}}}} \left(c_{\mathbf{p}} e^{-iE_{\mathbf{p}}t} + d^\dagger_{-\mathbf{p}} e^{iE_{\mathbf{p}}t} \right) \tag{44}$$

combining them, the mode expansion of the complex scalar field is

$$\phi(x) = \int d\Pi_p \left(c_{\mathbf{p}} e^{ip\cdot x} + d^\dagger_{\mathbf{p}} e^{-ip\cdot x} \right) \tag{45}$$

Where $d\Pi_p = \frac{d^3\mathbf{p}}{(2\pi)^{3/2}\sqrt{2E_{\mathbf{p}}}}$

For $\phi^\dagger$, the same procedure gives

$$\phi^\dagger(x) = \int d\Pi_p \left(\tilde{c}_{\mathbf{p}} e^{-ip\cdot x} + \tilde{d}^\dagger_{\mathbf{p}} e^{ip\cdot x} \right) \tag{46}$$

Since $\phi \neq \phi^\dagger$, $c_{\mathbf{p}}$ and $d_{\mathbf{p}}$ are independent, and the relation between $c_{\mathbf{p}}, d_{\mathbf{p}}$ and $\tilde{c}_{\mathbf{p}}, \tilde{d}_{\mathbf{p}}$ is determined by the hermitian conjugation:

$$\phi^\dagger(x) = \int d\Pi_p \left(c^\dagger_{\mathbf{p}} e^{-ip\cdot x} + d_{\mathbf{p}} e^{ip\cdot x} \right) \tag{47}$$

For the conjugate momenta, the mode expansion is

$$\pi(x) = \dot{\phi}^\dagger = \int d\Pi_p \left(iE_{\mathbf{p}} c^\dagger_{\mathbf{p}} e^{-ip\cdot x} - iE_{\mathbf{p}} d_{\mathbf{p}} e^{ip\cdot x} \right), \tag{48}$$

$$\pi^\dagger(x) = \dot{\phi} = \int d\Pi_p \left(-iE_{\mathbf{p}} c_{\mathbf{p}} e^{ip\cdot x} + iE_{\mathbf{p}} d^\dagger_{\mathbf{p}} e^{-ip\cdot x} \right) \tag{49}$$

The modes $c_{\mathbf{p}}$ and $d_{\mathbf{p}}$ also can be expressed in terms of the fields as follows:

$$E_{\mathbf{p}}\phi(x) + i\pi^\dagger(x) = \int d\Pi_q \left(E_{\mathbf{p}} c_{\mathbf{q}} e^{iq\cdot x} + E_{\mathbf{p}} d^\dagger_{\mathbf{q}} e^{-iq\cdot x} + E_{\mathbf{q}} c_{\mathbf{q}} e^{iq\cdot x} - E_{\mathbf{q}} d^\dagger_{\mathbf{q}} e^{-iq\cdot x} \right)$$

$$\implies \int d^3\mathbf{x}\, e^{-i\mathbf{p}\cdot\mathbf{x}} \left[E_{\mathbf{p}}\phi(x) + i\pi^\dagger(x) \right] = 2E_{\mathbf{p}} \int d\Pi_q \int d^3\mathbf{x}\, c_{\mathbf{q}} e^{i(\mathbf{q}-\mathbf{p})\cdot\mathbf{x}} e^{-iE_{\mathbf{q}}t} = \sqrt{2E_{\mathbf{p}}}(2\pi)^{3/2} e^{-iE_{\mathbf{p}}t} c_{\mathbf{p}}$$

$$\implies c_{\mathbf{p}} = \frac{e^{iE_{\mathbf{p}}t}}{(2\pi)^{3/2}\sqrt{2E_{\mathbf{p}}}} \int d^3\mathbf{x}\, e^{-i\mathbf{p}\cdot\mathbf{x}} \left[E_{\mathbf{p}}\phi(x) + i\pi^\dagger(x) \right] \tag{50}$$

and similarly,

$$E_{\mathbf{p}}\phi^\dagger(x) + i\pi(x) = \int d\Pi_q \left(E_{\mathbf{p}} c^\dagger_{\mathbf{q}} e^{-iq\cdot x} + E_{\mathbf{p}} d_{\mathbf{q}} e^{iq\cdot x} - E_{\mathbf{q}} c^\dagger_{\mathbf{q}} e^{-iq\cdot x} + E_{\mathbf{q}} d_{\mathbf{q}} e^{iq\cdot x}\right)$$

$$\implies \int d^3\mathbf{x}\, e^{-i\mathbf{p}\cdot\mathbf{x}} \left[E_{\mathbf{p}}\phi^\dagger(x) + i\pi(x)\right] = 2E_{\mathbf{p}} \int d\Pi_q \int d^3\mathbf{x}\, d_{\mathbf{q}} e^{i(\mathbf{q}-\mathbf{p})\cdot\mathbf{x}} e^{-iE_{\mathbf{q}}t} = \sqrt{2E_{\mathbf{p}}}(2\pi)^{3/2} e^{-iE_{\mathbf{p}}t} d_{\mathbf{p}}$$

$$\implies d_{\mathbf{p}} = \frac{e^{iE_{\mathbf{p}}t}}{(2\pi)^{3/2}\sqrt{2E_{\mathbf{p}}}} \int d^3\mathbf{x}\, e^{-i\mathbf{p}\cdot\mathbf{x}} \left[E_{\mathbf{p}}\phi^\dagger(x) + i\pi(x)\right] \tag{51}$$

Using these expressions, the commutation relations of $c_{\mathbf{p}}$ and $d_{\mathbf{p}}$ can be calculated from the canonical commutation relations of ϕ and π.

$$[E_{\mathbf{p}}\phi(x) + i\pi^\dagger(x), E_{\mathbf{q}}\phi(y) + i\pi^\dagger(y)] = 0 \implies [c_{\mathbf{p}}, c_{\mathbf{q}}] = 0 \tag{52}$$

$$[E_{\mathbf{p}}\phi^\dagger(x) + i\pi(x), E_{\mathbf{q}}\phi^\dagger(y) + i\pi(y)] = 0 \implies [d_{\mathbf{p}}, d_{\mathbf{q}}] = 0 \tag{53}$$

$$[E_{\mathbf{p}}\phi(x) + i\pi^\dagger(x), E_{\mathbf{q}}\phi^\dagger(y) + i\pi(y)] = (E_{\mathbf{q}} - E_{\mathbf{p}})\delta^{(3)}(\mathbf{x}-\mathbf{y}) \tag{54}$$

$$\int d^3\mathbf{x} d^3\mathbf{y}\, e^{-i\mathbf{p}\cdot\mathbf{x}} e^{-i\mathbf{q}\cdot\mathbf{y}} (E_{\mathbf{q}} - E_{\mathbf{p}})\delta^{(3)}(\mathbf{x}-\mathbf{y}) = \int d^3\mathbf{x}\, e^{-i(\mathbf{p}+\mathbf{q})\cdot\mathbf{x}}(E_{\mathbf{q}} - E_{\mathbf{p}}) = 0 \tag{55}$$

$$\implies [c_{\mathbf{p}}, d_{\mathbf{q}}] = 0 \tag{56}$$

$$[E_{\mathbf{p}}\phi(x) + i\pi^\dagger(x), E_{\mathbf{q}}\phi(y) - i\pi^\dagger(y)] = 0 \implies [c_{\mathbf{p}}, d^\dagger_{\mathbf{q}}] = 0 \tag{57}$$

$$[E_{\mathbf{p}}\phi(x) + i\pi^\dagger(x), E_{\mathbf{q}}\phi^\dagger(y) - i\pi(y)] = (E_{\mathbf{p}} + E_{\mathbf{q}})\delta^{(3)}(\mathbf{x}-\mathbf{y}) \tag{58}$$

$$\int d^3\mathbf{x} d^3\mathbf{y}\, e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{q}\cdot\mathbf{y}} (E_{\mathbf{p}} + E_{\mathbf{q}})\delta^{(3)}(\mathbf{x}-\mathbf{y}) = \int d^3\mathbf{x}\, e^{-i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}}(E_{\mathbf{p}} + E_{\mathbf{q}}) \tag{59}$$

$$= 2E_{\mathbf{p}}(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{60}$$

$$\implies [c_{\mathbf{p}}, c^\dagger_{\mathbf{q}}] = \delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{61}$$

$$[E_{\mathbf{p}}\phi^\dagger(x) + i\pi(x), E_{\mathbf{q}}\phi(y) - i\pi^\dagger(y)] = (E_{\mathbf{p}} + E_{\mathbf{q}})\delta^{(3)}(\mathbf{x}-\mathbf{y}) \tag{62}$$

$$\int d^3\mathbf{x} d^3\mathbf{y}\, e^{-i\mathbf{p}\cdot\mathbf{x}} e^{i\mathbf{q}\cdot\mathbf{y}} (E_{\mathbf{p}} + E_{\mathbf{q}})\delta^{(3)}(\mathbf{x}-\mathbf{y}) = \int d^3\mathbf{x}\, e^{-i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}}(E_{\mathbf{p}} + E_{\mathbf{q}}) \tag{63}$$

$$= 2E_{\mathbf{p}}(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{64}$$

$$\implies [d_{\mathbf{p}}, d^\dagger_{\mathbf{q}}] = \delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{65}$$

$E_{\mathbf{p}} = E_{-\mathbf{p}}$ is used to simplify the integrals. Hence, the commutation relations of $c_{\mathbf{p}}$ and $d_{\mathbf{p}}$ are

$$[c_{\mathbf{p}}, c_{\mathbf{q}}] = 0, \qquad [c_{\mathbf{p}}, c^\dagger_{\mathbf{q}}] = \delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{66}$$

$$[d_{\mathbf{p}}, d_{\mathbf{q}}] = 0, \qquad [d_{\mathbf{p}}, d^\dagger_{\mathbf{q}}] = \delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{67}$$

$$[c_{\mathbf{p}}, d_{\mathbf{q}}] = 0, \qquad [c_{\mathbf{p}}, d^\dagger_{\mathbf{q}}] = 0 \tag{68}$$

(c)

The Hamiltonian was

$$H = \int d^3\mathbf{x} \left(\pi^\dagger\pi + \nabla\phi^\dagger \cdot \nabla\phi + m^2\phi^\dagger\phi\right) \tag{69}$$

Each term are:

$$\int d^3\mathbf{x}\,\pi^\dagger\pi = \int d^3\mathbf{x}d\Pi_p d\Pi_q \left(-iE_\mathbf{p}c_\mathbf{p}e^{ip\cdot x} + iE_\mathbf{p}d^\dagger_\mathbf{p}e^{-ip\cdot x}\right)\left(iE_\mathbf{q}c^\dagger_\mathbf{q}e^{-iq\cdot x} - iE_\mathbf{q}d_\mathbf{q}e^{iq\cdot x}\right) \tag{70}$$

$$= \frac{1}{2(2\pi)^3}\int d^3\mathbf{p}d^3\mathbf{q}d^3\mathbf{x}\,\sqrt{E_\mathbf{p}E_\mathbf{q}}\left(c_\mathbf{p}e^{ip\cdot x} - d^\dagger_\mathbf{p}e^{-ip\cdot x}\right)\left(c^\dagger_\mathbf{q}e^{-iq\cdot x} - d_\mathbf{q}e^{iq\cdot x}\right) \tag{71}$$

$$= \frac{1}{2}\int d^3\mathbf{p}d^3\mathbf{q}\sqrt{E_\mathbf{p}E_\mathbf{q}}\delta^{(3)}(\mathbf{p}-\mathbf{q})\left[e^{-i(E_\mathbf{p}-E_\mathbf{q})t}c_\mathbf{p}c^\dagger_\mathbf{q} + e^{i(E_\mathbf{p}-E_\mathbf{q})t}d^\dagger_\mathbf{p}d_\mathbf{q}\right] \tag{72}$$

$$- \frac{1}{2}\int d^3\mathbf{p}d^3\mathbf{q}\sqrt{E_\mathbf{p}E_\mathbf{q}}\delta^{(3)}(\mathbf{p}+\mathbf{q})\left[e^{-i(E_\mathbf{p}+E_\mathbf{q})t}c_\mathbf{p}d_\mathbf{q} + e^{i(E_\mathbf{p}+E_\mathbf{q})t}d^\dagger_\mathbf{p}c^\dagger_\mathbf{q}\right] \tag{73}$$

$$= \frac{1}{2}\int d^3\mathbf{p}\,E_\mathbf{p}\left[c_\mathbf{p}c^\dagger_\mathbf{p} + d^\dagger_\mathbf{p}d_\mathbf{p} - e^{-2iE_\mathbf{p}t}c_\mathbf{p}d_{-\mathbf{p}} - e^{2iE_\mathbf{p}t}d^\dagger_\mathbf{p}c^\dagger_{-\mathbf{p}}\right] \tag{74}$$

and

$$\int d^3\mathbf{x}\,\phi^\dagger\phi = \int d^3\mathbf{x}d\Pi_p d\Pi_q \left(c^\dagger_\mathbf{p}e^{-ip\cdot x} + d_\mathbf{p}e^{ip\cdot x}\right)\left(c_\mathbf{q}e^{iq\cdot x} + d^\dagger_\mathbf{q}e^{-iq\cdot x}\right) \tag{75}$$

$$= \frac{1}{2(2\pi)^3}\int d^3\mathbf{p}d^3\mathbf{q}d^3\mathbf{x}\,\frac{1}{\sqrt{E_\mathbf{p}E_\mathbf{q}}}\left(c^\dagger_\mathbf{p}e^{-ip\cdot x} + d_\mathbf{p}e^{ip\cdot x}\right)\left(c_\mathbf{q}e^{iq\cdot x} + d^\dagger_\mathbf{q}e^{-iq\cdot x}\right) \tag{76}$$

$$= \frac{1}{2}\int d^3\mathbf{p}d^3\mathbf{q}\frac{1}{\sqrt{E_\mathbf{p}E_\mathbf{q}}}\delta^{(3)}(\mathbf{p}-\mathbf{q})\left[e^{i(E_\mathbf{p}-E_\mathbf{q})t}c^\dagger_\mathbf{p}c_\mathbf{q} + e^{-i(E_\mathbf{p}-E_\mathbf{q})t}d_\mathbf{p}d^\dagger_\mathbf{q}\right] \tag{77}$$

$$+ \frac{1}{2}\int d^3\mathbf{p}d^3\mathbf{q}\frac{1}{\sqrt{E_\mathbf{p}E_\mathbf{q}}}\delta^{(3)}(\mathbf{p}+\mathbf{q})\left[e^{i(E_\mathbf{p}+E_\mathbf{q})t}c^\dagger_\mathbf{p}d^\dagger_\mathbf{q} + e^{-i(E_\mathbf{p}+E_\mathbf{q})t}d_\mathbf{p}c_\mathbf{q}\right] \tag{78}$$

$$= \frac{1}{2}\int d^3\mathbf{p}\,\frac{1}{E_\mathbf{p}}\left[c^\dagger_\mathbf{p}c_\mathbf{p} + d_\mathbf{p}d^\dagger_\mathbf{p} + e^{2iE_\mathbf{p}t}c^\dagger_\mathbf{p}d^\dagger_{-\mathbf{p}} + e^{-2iE_\mathbf{p}t}d_\mathbf{p}c_{-\mathbf{p}}\right] \tag{79}$$

And the spatial derivatives term is:

$$\int d^3\mathbf{x}\,\nabla\phi^\dagger\cdot\nabla\phi = \int d^3\mathbf{x}d\Pi_p d\Pi_q \left(-i\mathbf{p}c^\dagger_\mathbf{p}e^{-ip\cdot x} + i\mathbf{p}d_\mathbf{p}e^{ip\cdot x}\right)\cdot\left(i\mathbf{q}c_\mathbf{q}e^{iq\cdot x} - i\mathbf{q}d^\dagger_\mathbf{q}e^{-iq\cdot x}\right) \tag{80}$$

$$= \frac{1}{2(2\pi)^3}\int d^3\mathbf{p}d^3\mathbf{q}d^3\mathbf{x}\,\frac{\mathbf{p}\cdot\mathbf{q}}{\sqrt{E_\mathbf{p}E_\mathbf{q}}}\left(c^\dagger_\mathbf{p}e^{-ip\cdot x} - d_\mathbf{p}e^{ip\cdot x}\right)\left(c_\mathbf{q}e^{iq\cdot x} - d^\dagger_\mathbf{q}e^{-iq\cdot x}\right) \tag{81}$$

$$= \frac{1}{2}\int d^3\mathbf{p}d^3\mathbf{q}\frac{\mathbf{p}\cdot\mathbf{q}}{\sqrt{E_\mathbf{p}E_\mathbf{q}}}\delta^{(3)}(\mathbf{p}-\mathbf{q})\left[e^{i(E_\mathbf{p}-E_\mathbf{q})t}c^\dagger_\mathbf{p}c_\mathbf{q} + e^{-i(E_\mathbf{p}-E_\mathbf{q})t}d_\mathbf{p}d^\dagger_\mathbf{q}\right] \tag{82}$$

$$- \frac{1}{2}\int d^3\mathbf{p}d^3\mathbf{q}\frac{\mathbf{p}\cdot\mathbf{q}}{\sqrt{E_\mathbf{p}E_\mathbf{q}}}\delta^{(3)}(\mathbf{p}+\mathbf{q})\left[e^{i(E_\mathbf{p}+E_\mathbf{q})t}c^\dagger_\mathbf{p}d^\dagger_\mathbf{q} + e^{-i(E_\mathbf{p}+E_\mathbf{q})t}d_\mathbf{p}c_\mathbf{q}\right] \tag{83}$$

$$= \frac{1}{2}\int d^3\mathbf{p}\,\frac{|\mathbf{p}|^2}{E_\mathbf{p}}\left[c^\dagger_\mathbf{p}c_\mathbf{p} + d_\mathbf{p}d^\dagger_\mathbf{p} + e^{2iE_\mathbf{p}t}c^\dagger_\mathbf{p}d^\dagger_{-\mathbf{p}} + e^{-2iE_\mathbf{p}t}d_\mathbf{p}c_{-\mathbf{p}}\right] \tag{84}$$

using the relation $E^2_\mathbf{p} = |\mathbf{p}|^2 + m^2$,

$$\int d^3\mathbf{x}\,\left(\nabla\phi^\dagger\cdot\nabla\phi + m^2\phi^\dagger\phi\right) = \frac{1}{2}\int d^3\mathbf{p}\,E_\mathbf{p}\left[c^\dagger_\mathbf{p}c_\mathbf{p} + d_\mathbf{p}d^\dagger_\mathbf{p} + e^{2iE_\mathbf{p}t}c^\dagger_\mathbf{p}d^\dagger_{-\mathbf{p}} + e^{-2iE_\mathbf{p}t}d_\mathbf{p}c_{-\mathbf{p}}\right] \tag{85}$$

To cancel the last two terms, change the variable $\mathbf{p}\to-\mathbf{p}$ and use the commutation relations $[c_\mathbf{p}, d_\mathbf{q}] = [c^\dagger_\mathbf{p}, d^\dagger_\mathbf{q}] = 0$. Since both $|p|^2$ and $E_\mathbf{p}$ are invariant under $\mathbf{p}\to-\mathbf{p}$,

$$\int d^3\mathbf{x}\,\left(\nabla\phi^\dagger\cdot\nabla\phi + m^2\phi^\dagger\phi\right) = \frac{1}{2}\int d^3\mathbf{p}\,E_\mathbf{p}\left[c^\dagger_\mathbf{p}c_\mathbf{p} + d_\mathbf{p}d^\dagger_\mathbf{p} + e^{-2iE_\mathbf{p}t}c_\mathbf{p}d_{-\mathbf{p}} + e^{2iE_\mathbf{p}t}d^\dagger_\mathbf{p}c^\dagger_{-\mathbf{p}}\right] \tag{86}$$

Hence, the Hamiltonian is

$$H = \frac{1}{2}\int d^3\mathbf{p}\,E_\mathbf{p}\left(c^\dagger_\mathbf{p}c_\mathbf{p} + c_\mathbf{p}c^\dagger_\mathbf{p} + d^\dagger_\mathbf{p}d_\mathbf{p} + d_\mathbf{p}d^\dagger_\mathbf{p}\right) \tag{87}$$

Since exchanging the order of $c_{\mathbf{p}}$ and $c_{\mathbf{p}}^\dagger$ gives only a constant (which is the vacuum energy contribution, $\delta^{(3)}(0)\int d^3\mathbf{p}E_{\mathbf{p}}$), the equivalent Hamiltonian can be written as

$$H = \int d^3\mathbf{p}\, E_{\mathbf{p}} \left(c_{\mathbf{p}}^\dagger c_{\mathbf{p}} + d_{\mathbf{p}}^\dagger d_{\mathbf{p}}\right) \tag{88}$$

Different normalization of mode ($c_{\mathbf{p}} = (2\pi)^{-3/2}a_{\mathbf{p}}, d_{\mathbf{p}} = (2\pi)^{-3/2}b_{\mathbf{p}}$) varies the hamiltonian and commutation relation upto a constant:

$$H = \int \frac{d^3\mathbf{p}}{(2\pi)^3}\, E_{\mathbf{p}} \left(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + b_{\mathbf{p}}^\dagger b_{\mathbf{p}}\right), \qquad [a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = [b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{89}$$

since $[H, a_{\mathbf{p}}^\dagger] = E_{\mathbf{p}}a_{\mathbf{p}}^\dagger$ and $[H, b_{\mathbf{p}}^\dagger] = E_{\mathbf{p}}b_{\mathbf{p}}^\dagger$, both $a_{\mathbf{p}}^\dagger$ and $b_{\mathbf{p}}^\dagger$ create a particle with energy $E_{\mathbf{p}}$. And since $[a_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = 0$ and $\langle 0|b_{\mathbf{p}}^\dagger a_{\mathbf{q}}|0\rangle$ they are independent species of particles.

(d)

The invariance under the transformation $\phi \to \phi e^{i\alpha} = \phi + i\alpha\phi + O(\alpha^2)$ and $\phi^\dagger \to \phi^\dagger e^{-i\alpha} = \phi^\dagger - i\alpha\phi^\dagger + O(\alpha^2)$ gives the conserved charge

$$i\int d^3\mathbf{x} \left(\phi\pi - \phi^\dagger\pi^\dagger\right) \tag{90}$$

Expanding the fields in terms of the modes, the normal ordered charge upto multipliers is

$$Q \sim \int \frac{d^3\mathbf{p}}{(2\pi)^3} \left(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}}^\dagger b_{\mathbf{p}}\right) \tag{91}$$

Since $[a_{\mathbf{p}}, b_{\mathbf{q}}] = [a_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = 0$ and $[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = [b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})$, the commutation relations of the number operators are

$$[a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, b_{\mathbf{q}}^\dagger b_{\mathbf{q}}] = a_{\mathbf{p}}^\dagger b_{\mathbf{q}}^\dagger [a_{\mathbf{p}}, b_{\mathbf{q}}] + a_{\mathbf{p}}^\dagger [a_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] b_{\mathbf{q}} + b_{\mathbf{q}}^\dagger [a_{\mathbf{p}}^\dagger, b_{\mathbf{q}}] a_{\mathbf{p}} + [a_{\mathbf{p}}^\dagger, b_{\mathbf{q}}^\dagger] b_{\mathbf{q}} a_{\mathbf{p}} = 0 \tag{92}$$

and

$$[a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger a_{\mathbf{q}}] = a_{\mathbf{p}}^\dagger a_{\mathbf{q}}^\dagger [a_{\mathbf{p}}, a_{\mathbf{q}}] + a_{\mathbf{p}}^\dagger [a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] a_{\mathbf{q}} + a_{\mathbf{q}}^\dagger [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}] a_{\mathbf{p}} + [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger] a_{\mathbf{q}} a_{\mathbf{p}} \tag{93}$$

$$= (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) \left(a_{\mathbf{p}}^\dagger a_{\mathbf{q}} - a_{\mathbf{q}}^\dagger a_{\mathbf{p}}\right) = 0 \tag{94}$$

similarly, $[b_{\mathbf{p}}^\dagger b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger b_{\mathbf{q}}] = 0$.

Hence, conseration of the charge is verified by the mode expansion of the Hamiltonian and the charge:

$$[H, Q] = \int \frac{d^3\mathbf{p}}{(2\pi)^3}\frac{d^3\mathbf{q}}{(2\pi)^3} E_{\mathbf{p}} \left([a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger a_{\mathbf{q}}] + [b_{\mathbf{p}}^\dagger b_{\mathbf{p}}, a_{\mathbf{q}}^\dagger a_{\mathbf{q}}] - [a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, b_{\mathbf{q}}^\dagger b_{\mathbf{q}}] - [b_{\mathbf{p}}^\dagger b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger b_{\mathbf{q}}]\right) = 0 \tag{95}$$

Recall the commutation relations with the number operator:

$$[a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] = a_{\mathbf{p}}^\dagger [a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger] + [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger] a_{\mathbf{p}} = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) a_{\mathbf{p}}^\dagger, \tag{96}$$

$$[a_{\mathbf{p}}^\dagger a_{\mathbf{p}}, a_{\mathbf{q}}] = a_{\mathbf{p}}^\dagger [a_{\mathbf{p}}, a_{\mathbf{q}}] + [a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}] a_{\mathbf{p}} = -(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) a_{\mathbf{p}}, \tag{97}$$

$$[b_{\mathbf{p}}^\dagger b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] = b_{\mathbf{p}}^\dagger [b_{\mathbf{p}}, b_{\mathbf{q}}^\dagger] + [b_{\mathbf{p}}^\dagger, b_{\mathbf{q}}^\dagger] b_{\mathbf{p}} = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) b_{\mathbf{p}}^\dagger, \tag{98}$$

$$[b_{\mathbf{p}}^\dagger b_{\mathbf{p}}, b_{\mathbf{q}}] = b_{\mathbf{p}}^\dagger [b_{\mathbf{p}}, b_{\mathbf{q}}] + [b_{\mathbf{p}}^\dagger, b_{\mathbf{q}}] b_{\mathbf{p}} = -(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) b_{\mathbf{p}} \tag{99}$$

Hence,

$$[Q, a_{\mathbf{p}}] = \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left([a^\dagger_{\mathbf{q}} a_{\mathbf{q}}, a_{\mathbf{p}}] - [b^\dagger_{\mathbf{q}} b_{\mathbf{q}}, a_{\mathbf{p}}] \right) = -a_{\mathbf{p}}, \tag{100}$$

$$[Q, a^\dagger_{\mathbf{p}}] = \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left([a^\dagger_{\mathbf{q}} a_{\mathbf{q}}, a^\dagger_{\mathbf{p}}] - [b^\dagger_{\mathbf{q}} b_{\mathbf{q}}, a^\dagger_{\mathbf{p}}] \right) = a^\dagger_{\mathbf{p}}, \tag{101}$$

$$[Q, b_{\mathbf{p}}] = \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left([a^\dagger_{\mathbf{q}} a_{\mathbf{q}}, b_{\mathbf{p}}] - [b^\dagger_{\mathbf{q}} b_{\mathbf{q}}, b_{\mathbf{p}}] \right) = b_{\mathbf{p}}, \tag{102}$$

$$[Q, b^\dagger_{\mathbf{p}}] = \int \frac{d^3\mathbf{q}}{(2\pi)^3} \left([a^\dagger_{\mathbf{q}} a_{\mathbf{q}}, b^\dagger_{\mathbf{p}}] - [b^\dagger_{\mathbf{q}} b_{\mathbf{q}}, b^\dagger_{\mathbf{p}}] \right) = -b^\dagger_{\mathbf{p}} \tag{103}$$

and the U(1) quantum number of $a^\dagger_{\mathbf{p}} |0\rangle$ is 1 and that of $b^\dagger_{\mathbf{p}} |0\rangle$ is -1.

Therefore, the commutator with the field is

$$[Q, \phi(x)] = \int \frac{d^3\mathbf{p}}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left([Q, a_{\mathbf{p}}] e^{ip\cdot x} + [Q, b^\dagger_{\mathbf{p}}] e^{-ip\cdot x} \right) \tag{104}$$

$$= \int \frac{d^3\mathbf{p}}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left(-a_{\mathbf{p}} e^{ip\cdot x} - b^\dagger_{\mathbf{p}} e^{-ip\cdot x} \right) \tag{105}$$

$$= -\phi(x) \tag{106}$$

$$[Q, \phi^\dagger(x)] = \int \frac{d^3\mathbf{p}}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left([Q, a^\dagger_{\mathbf{p}}] e^{-ip\cdot x} + [Q, b_{\mathbf{p}}] e^{ip\cdot x} \right) \tag{107}$$

$$= \int \frac{d^3\mathbf{p}}{(2\pi)^3 \sqrt{2E_{\mathbf{p}}}} \left(a^\dagger_{\mathbf{p}} e^{-ip\cdot x} + b_{\mathbf{p}} e^{ip\cdot x} \right) \tag{108}$$

$$= \phi^\dagger(x) \tag{109}$$

For the composite operators,

$$[Q, \phi(x)\phi(y)] = [Q, \phi(x)]\phi(y) + \phi(x)[Q, \phi(y)] = -\phi(x)\phi(y) - \phi(x)\phi(y) = -2\phi(x)\phi(y) \tag{110}$$

$$[Q, \phi^\dagger(x)\phi(y)] = [Q, \phi^\dagger(x)]\phi(y) + \phi^\dagger(x)[Q, \phi(y)] = \phi^\dagger(x)\phi(y) - \phi^\dagger(x)\phi(y) = 0 \tag{111}$$

Therefore, the U(1) quantum number of $\phi\phi$ is twice that of ϕ, and the U(1) quantum number of $\phi^\dagger\phi$ is zero.

Problem 3.3

(a)

The spatial fourier transform of Green's function is

$$G(0,\mathbf{x}) = \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot\mathbf{x}} g_{\mathbf{p}}(0) \tag{112}$$

Since $g_{\mathbf{p}}(t)$ is continuous at $t=0$, the boundary condition of retarded and advanced Green's function implies $g_{\mathbf{p}}(0)=0$. Hence,

$$G_R(0,\mathbf{x}) = G_A(0,\mathbf{x}) = 0 \tag{113}$$

To extend this result to all space-like points, we need $\mathrm{SO}^+(1,3)$ invariance of the Green's functions. It can be shown by the momentum-space representation:

$$G_{R/A}(x) = \int \frac{d^4p}{(2\pi)^4} e^{ip\cdot x} \tilde{G}_{R/A}(p), \tag{114}$$

$$\widetilde{G}_R(p) = \lim_{\varepsilon\to 0^+} \frac{i}{(p^0+i\varepsilon)^2 - E_{\mathbf{p}}^2}, \tag{115}$$

$$\widetilde{G}_A(p) = \lim_{\varepsilon\to 0^+} \frac{i}{(p^0-i\varepsilon)^2 - E_{\mathbf{p}}^2}, \tag{116}$$

Even though above expression is not manifestly $\mathrm{SO}(1,3)$ invariant, it can be transformed into an invariant form as follows:

$$\widetilde{G}_R(p) = \lim_{\varepsilon\to 0^+} \frac{i}{(p^0+i\varepsilon)^2 - E_{\mathbf{p}}^2} \tag{117}$$

$$= \lim_{\varepsilon\to 0^+} \frac{i}{(p^0)^2 + 2i\varepsilon p^0 - \varepsilon^2 - E_{\mathbf{p}}^2} \tag{118}$$

$$= \lim_{\varepsilon\to 0^+} \frac{i}{-p^2 - m^2 + 2i\varepsilon p^0 - \varepsilon^2} \tag{119}$$

$$= \lim_{\varepsilon\to 0^+} \frac{i}{-p^2 - m^2 + 2i\varepsilon p^0} \tag{120}$$

$$= \lim_{\eta\to 0^+} \frac{i}{-p^2 - m^2 + i\eta \operatorname{sgn} p^0} \qquad (\eta = 2\varepsilon|p^0|) \tag{121}$$

Both p^2 and m^2 are $\mathrm{SO}^+(1,3)$ invariant and $\operatorname{sgn} p^0$ is also $\mathrm{SO}^+(1,3)$ invariant for time-like p. This is enough because the limit $\eta\to 0^+$ only meaningful for p near the pole $p^2=-m^2$ and this means p is time-like. Therefore, the retarded Green's function is $\mathrm{SO}^+(1,3)$ invariant. Similarly, the advanced Green's function is also $\mathrm{SO}^+(1,3)$ invariant:

$$\widetilde{G}_A(p) = \lim_{\eta\to 0^+} \frac{i}{-p^2 - m^2 - i\eta \operatorname{sgn} p^0} \tag{122}$$

Hence, the position-space Green functions are also $\mathrm{SO}^+(1,3)$ invariant and all space-like points can be transformed to $(0,\mathbf{x})$ by a orthochronous Lorentz transformation. Therefore,

$$G_R(x) = G_A(x) = 0 \qquad \text{for all } x^2 > 0 \tag{123}$$

For the Feynman Green's function, following position-space representation is the counterexample.

$$G_F(x) = \frac{m}{4\pi^2\sqrt{x^2}} K_1\left(m\sqrt{x^2}\right), \tag{124}$$

(b)

Start from the spatial Fourier transform of the Green's function:

$$g_{R,\mathbf{p}}(t) = -i\Theta(t)\frac{\sin E_{\mathbf{p}}t}{E_{\mathbf{p}}}, \qquad g_{A,\mathbf{p}}(t) = i\Theta(-t)\frac{\sin E_{\mathbf{p}}t}{E_{\mathbf{p}}} \tag{125}$$

and the two-point function $W(x-y) = \langle 0|\phi(x)\phi(y)|0\rangle$:

$$W(x) = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2E_{\mathbf{p}}} e^{ip\cdot x} \tag{126}$$

$$\langle 0|[\phi(x),\phi(y)]|0\rangle = W(x-y) - W(y-x) \tag{127}$$

$$= \int \frac{d^3\mathbf{p}}{(2\pi)^3 2E_{\mathbf{p}}} \left(e^{ip\cdot(x-y)} - e^{-ip\cdot(x-y)}\right) \tag{128}$$

$$= \int \frac{d^3\mathbf{p}}{(2\pi)^3 2E_{\mathbf{p}}} \left(e^{-iE_{\mathbf{p}}t}e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{iE_{\mathbf{p}}t}e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\right) \tag{129}$$

Combining the retarded and advanced Green's functions, we have

$$g_{R,\mathbf{p}}(t) - g_{A,\mathbf{p}}(t) = -i\frac{\sin E_{\mathbf{p}}t}{E_{\mathbf{p}}} \tag{130}$$

To express the expectation of commutator in terms of the retarded and advanced Green's functions, invert the three-momentum using the fact that $E_{\mathbf{p}} = E_{-\mathbf{p}}$ and the measure is invariant under $\mathbf{p} \to -\mathbf{p}$:

$$\langle 0|[\phi(x),\phi(y)]|0\rangle = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2E_{\mathbf{p}}} \left(e^{-iE_{\mathbf{p}}t}e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} - e^{iE_{\mathbf{p}}t}e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\right) \tag{131}$$

$$= i\int \frac{d^3\mathbf{p}}{(2\pi)^3 E_{\mathbf{p}}} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \frac{\left(e^{-iE_{\mathbf{p}}t} - e^{iE_{\mathbf{p}}t}\right)}{2i} \tag{132}$$

$$= -i\int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \frac{\sin E_{\mathbf{p}}t}{E_{\mathbf{p}}} \tag{133}$$

$$= \int \frac{d^3\mathbf{p}}{(2\pi)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} \left(g_{R,\mathbf{p}}(t) - g_{A,\mathbf{p}}(t)\right) \tag{134}$$

$$= G_R(x-y) - G_A(x-y) \tag{135}$$

Where $t = x^0 - y^0$.

For space-likely separated points x and y,

$$G_R(x-y) - G_A(x-y) = 0 \implies \langle 0|[\phi(x),\phi(y)]|0\rangle = 0 \tag{136}$$

Since the commutator is a c-number (Equation. 14), this can be generalized to operator identity, which is the microcausality:

$$[\phi(x),\phi(y)] = 0 \qquad \text{for all } (x-y)^2 > 0 \tag{137}$$